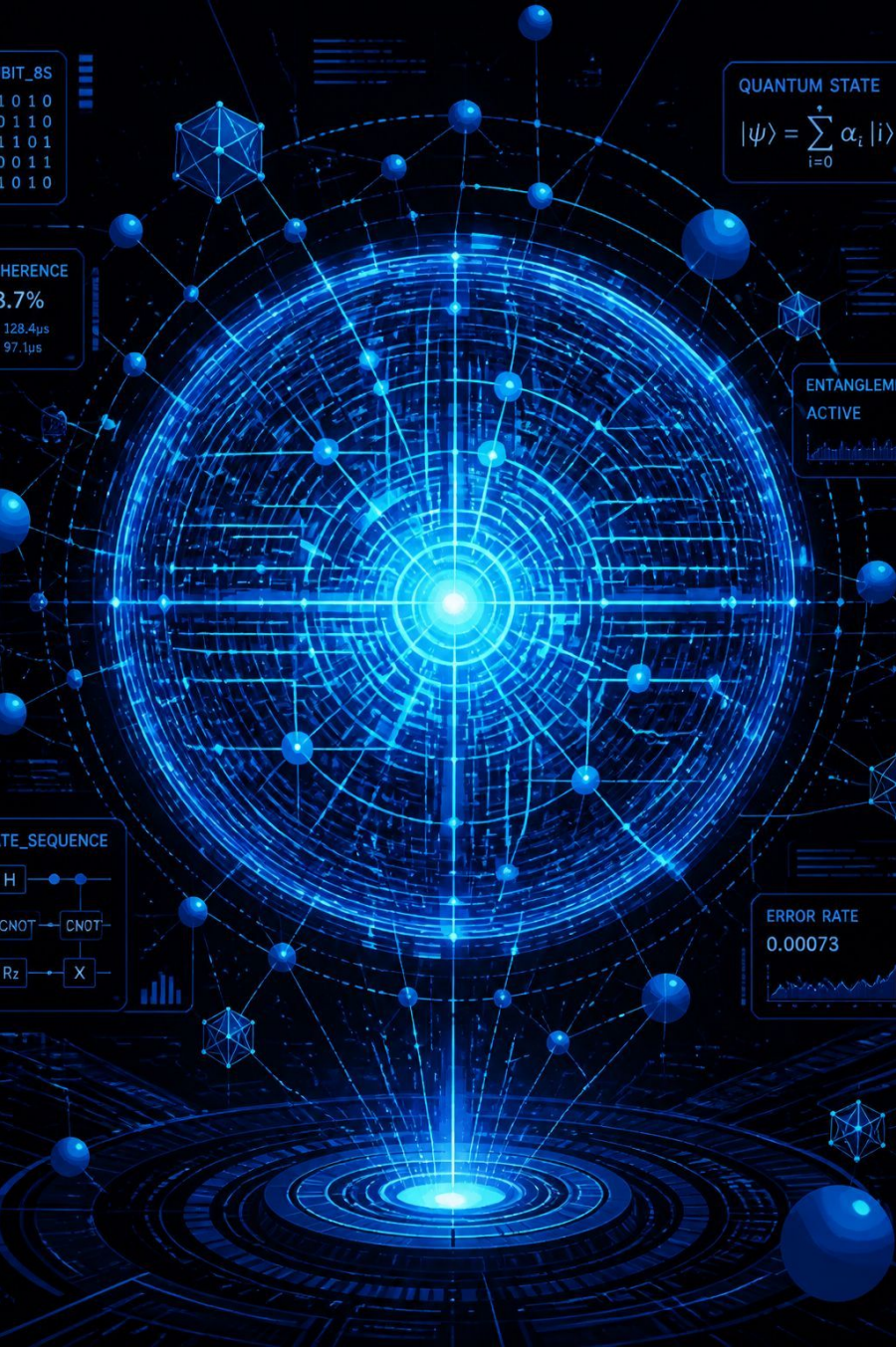




Quantum Computation: History & Overview

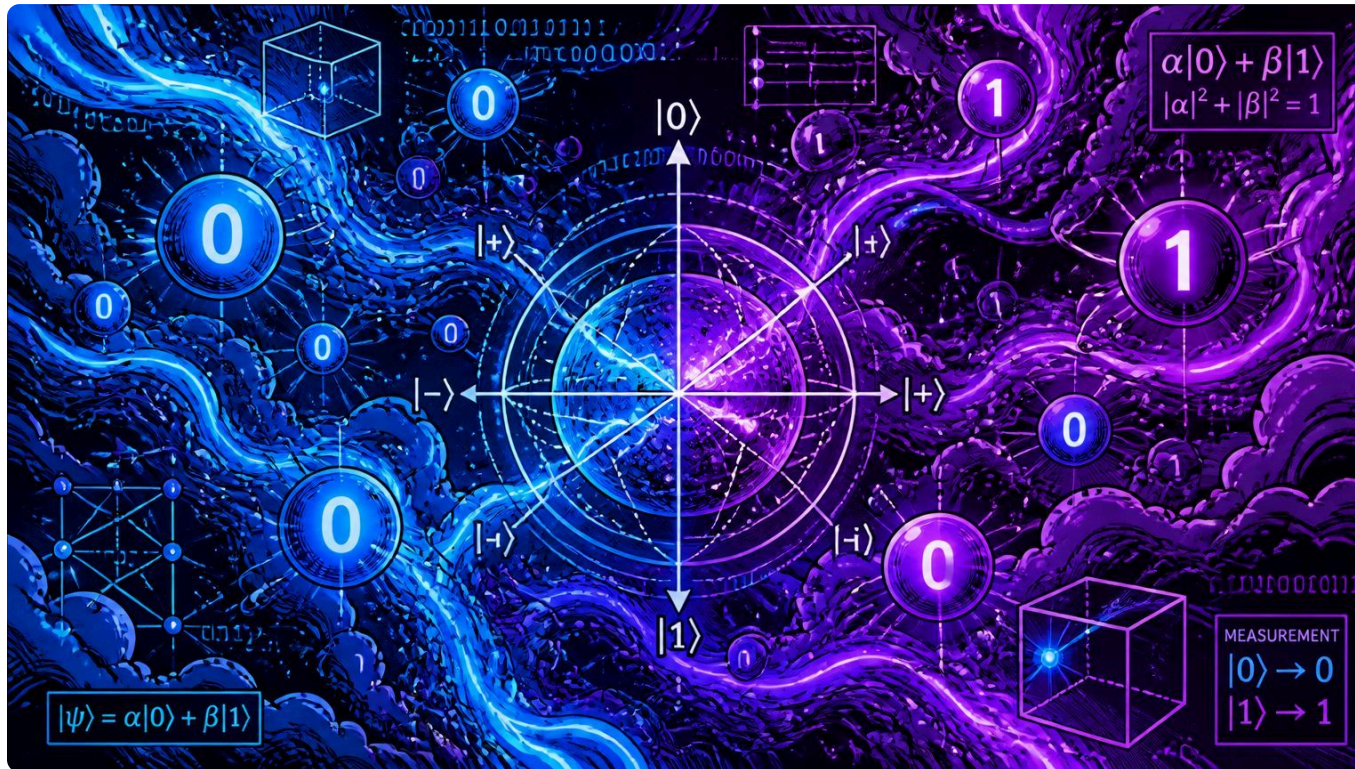
A journey through one of the most transformative ideas in modern science
— from Planck's quantum theory to Google's quantum supremacy.

COMPUTER SCIENCE

QUANTUM MECHANICS

TECHNOLOGY

What Is Quantum Computation?



Quantum computation is a computing paradigm grounded in the principles of **quantum mechanics**. Instead of classical bits, it uses **qubits** — which can exist as 0, 1, or *both simultaneously* through superposition.

This enables quantum computers to explore many solutions in parallel, offering exponential speedups for certain problem classes.

Cryptography

Drug Discovery

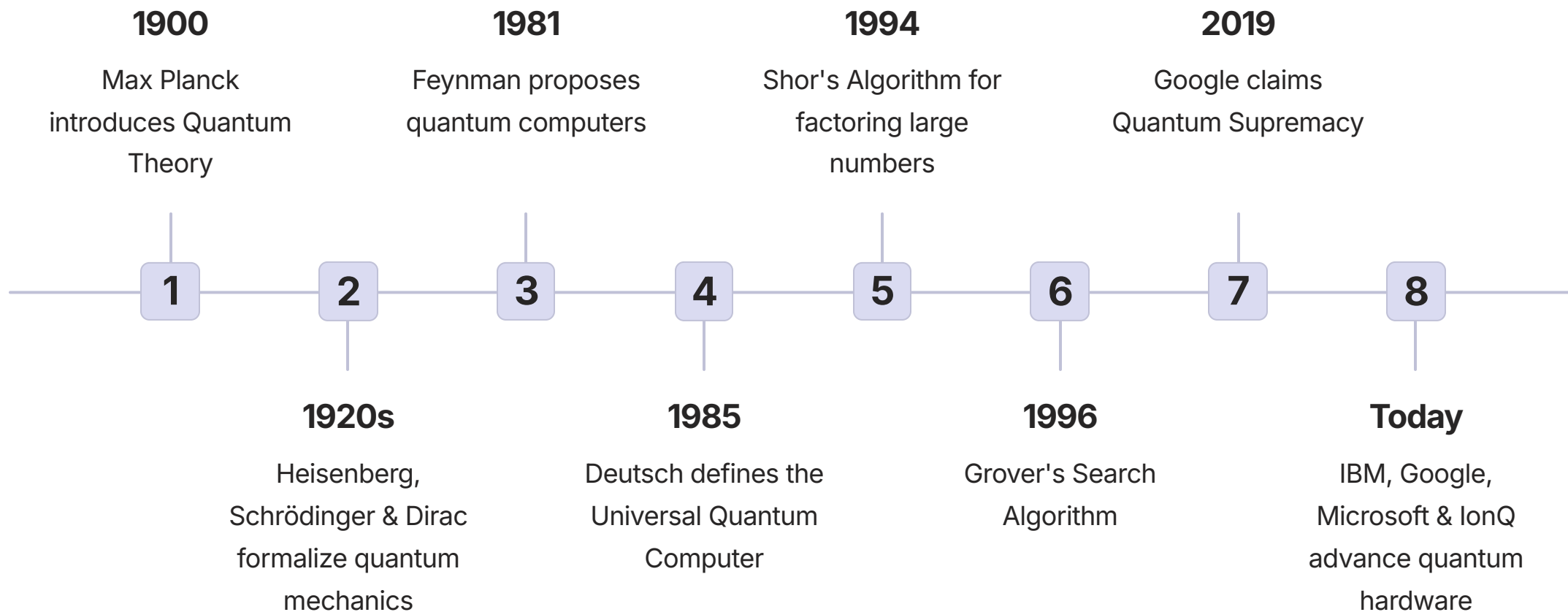
Optimization

AI & ML

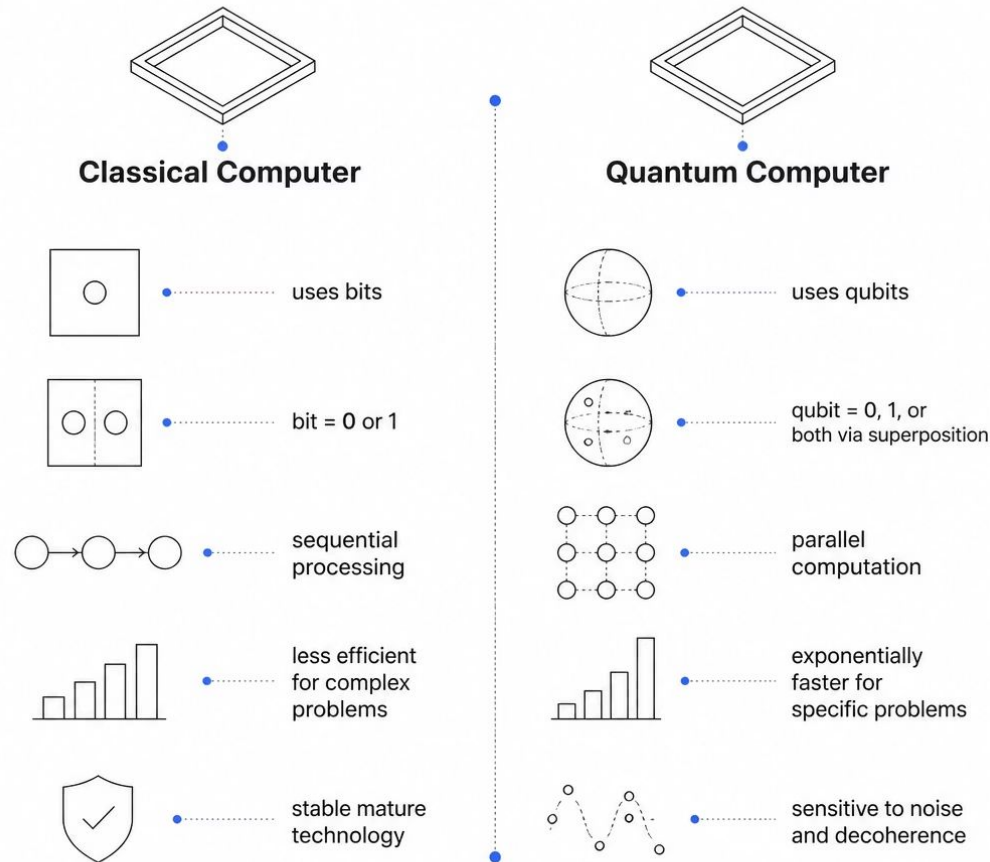
Material
Science

Milestones in Quantum Computing

From theory to reality — over a century of breakthroughs built the foundation for quantum computing.



Classical vs. Quantum Computing



The Fundamental Difference

Classical computers encode information as bits — strictly 0 *or* 1. Quantum computers use qubits, which exploit superposition to represent **both states at once**.

This allows quantum machines to evaluate many possibilities simultaneously, making them dramatically faster for problems like prime factorization, database search, and molecular simulation.

📄 A quantum computer with just 300 qubits could represent more states than there are atoms in the observable universe.

The Four Pillars of Quantum Computing

Qubits

The fundamental unit of quantum information — the quantum analog of the classical bit.

Superposition

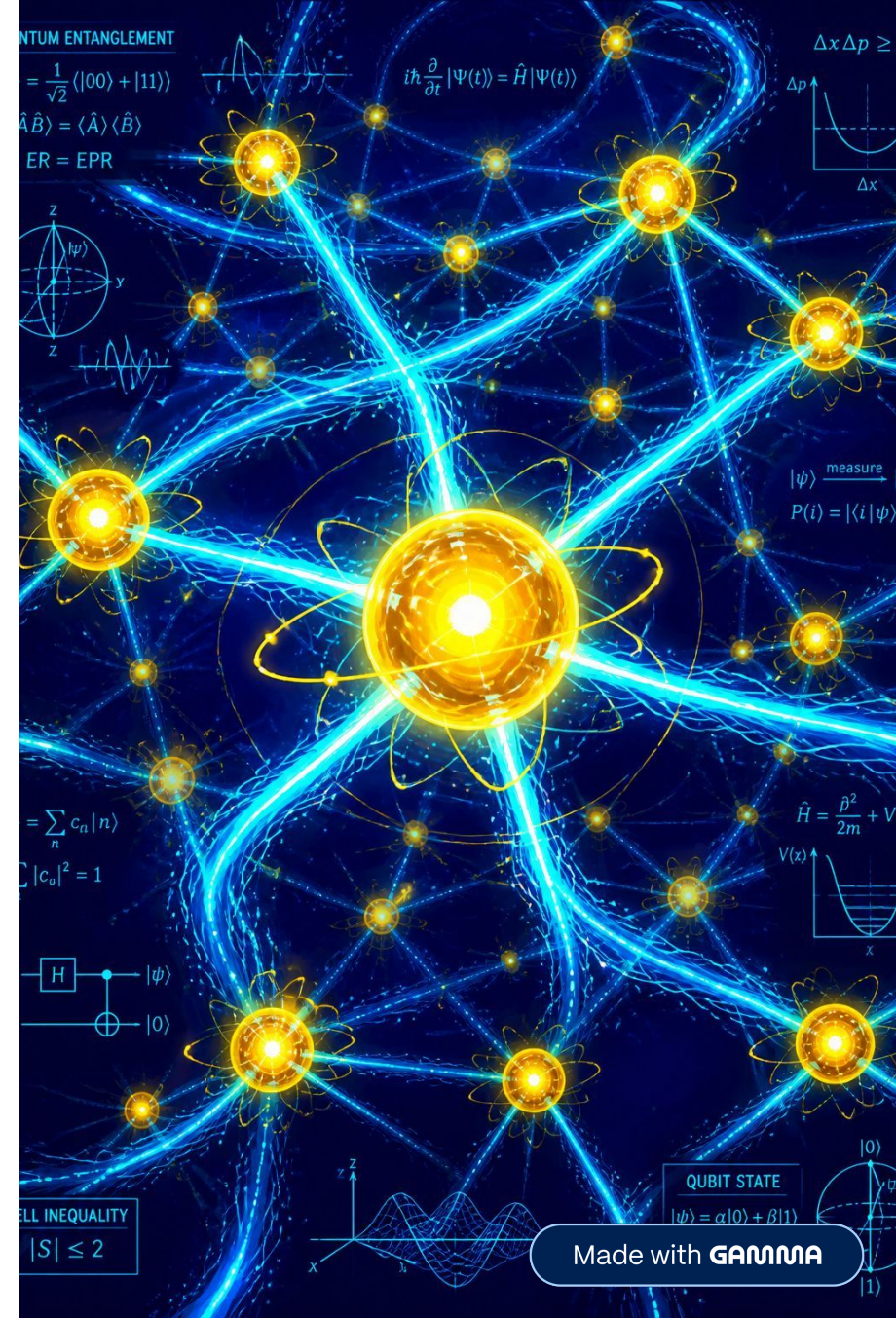
A qubit exists in multiple states simultaneously until measured, enabling parallel computation.

Entanglement

Two qubits become intrinsically linked — measuring one instantly determines the other, regardless of distance.

Interference

Quantum states amplify correct answers and cancel incorrect ones, steering computation toward the right solution.



Inside a Quantum Computer



A quantum computer is a complex stack of specialized components working in concert to maintain and control fragile quantum states.

→ **Quantum Processor (QPU)**

Houses the qubits and performs quantum operations

→ **Quantum Gates**

Manipulate qubit states, analogous to logic gates

→ **Cryogenic Cooling**

Maintains near-absolute-zero temperatures (~15 mK)

→ **Control Electronics & Classical Interface**

Translates classical instructions into quantum operations

Why Quantum Computing Is a Game-Changer

2^n

Parallel States

n qubits represent 2^n states simultaneously

$\sim 15\text{mK}$

Operating Temp

Colder than outer space for qubit stability

2019

Supremacy Year

Google's Sycamore solved a task in 200 seconds

1000+

Qubit Roadmap

IBM's target for fault-tolerant systems

Quantum computers offer **exponential speedups** for optimization, molecular simulation, and database search — with transformative potential for AI and scientific discovery.

The Hard Problems We Must Solve

Quantum computing faces profound engineering hurdles before it can scale reliably.

Decoherence

Qubits lose their quantum state through environmental interaction, causing computation errors.

Error Rates

Current qubits have high error rates, requiring complex error-correction overhead.

Hardware Cost

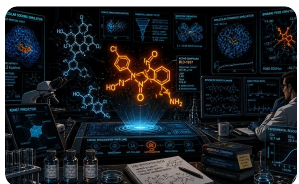
Cryogenic systems and precision control electronics are extremely expensive.

Scalability

Building systems with thousands of reliable, interconnected qubits remains an open challenge.

Real-World Applications

Quantum computing is poised to disrupt industries where classical computers hit fundamental limits.



Drug Discovery

Simulate molecular interactions at quantum precision, accelerating pharmaceutical development.



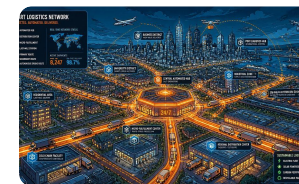
Financial Modeling

Optimize portfolios, detect fraud, and model risk with unprecedented speed.



Cybersecurity

Break current encryption (Shor's algorithm) and enable quantum-safe communication.



Logistics & Optimization

Solve complex routing and scheduling problems that are intractable classically.



The Road Ahead

Quantum computing is still in its infancy — but the roadmap is ambitious and the potential is enormous.

Fault-Tolerant Systems

Error-corrected qubits enabling reliable, large-scale computation

Quantum Internet

Entanglement-based networks for ultra-secure communication

Quantum Cloud

Commercial access to quantum processors via cloud platforms

Advanced Algorithms

New algorithms unlocking applications we can't yet imagine

- ❏ **Key Takeaway:** Quantum computing harnesses superposition, entanglement, and interference to solve problems beyond classical reach — but significant engineering challenges remain before widespread adoption.